2025 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)
29-30 November 2025, Eastern University, Dhaka, Bangladesh

2025 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON) | 979-8-3315-6105-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/BECITHCON69222.2025.11504114

A Comparative Analysis of Machine Learning Algorithms for the Identification of Parkinson's Disease Using Optimized Feature Selection

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***Abstract*— In order to predict Parkinson's disease, which impacts both motor and non-motor systems, including early stage speech loss, this study evaluates the use of voice based biomarkers using speech recordings. The purpose of this research is to examine the various machine learning techniques for early Parkinson's disease prediction. The dataset is available in the UCI Machine Learning repository. The dataset has 195 data points. To identify the most essential characteristics, Recursive Feature Elimination (RFE) was applied as feature selection methods. The top 10 features chosen by RFE were utilized. The performance of several machine learning techniques, including SVM, KNN, RF, DT, NB, LR, AdaBoost, CatBoost, and Gradient Boosting, was investigated deeper after being applied to this dataset. To maximize the performance, predictions from several models were tuned by hyperparameter tuning, which achieved an accuracy of 95.79% using CatBoost.**

***Keywords*— Parkinson's disease, Machine learning (ML), MDVP, Feature selection, KNN, RF, Hyperparameter tuning, Early detection, SMOTE, Vocal biomarkers.**

I. INTRODUCTION

Over many years, Parkinson's disease (PD) causes gradual harm to several areas of the brain. Millions of individuals worldwide suffer from this progressive neurological illness. PD is the second most common neurological disorder after Alzheimer's disease. James Parkinson, a British scientist discovered PD tremor palsy in 1817 and described its signs [1]. Dopamine is a neurotransmitter that serves as the primary chemical messenger for bodily movement [2]. The general functioning of the organism is thus adversely affected by any injury or death to the neurons that produce dopamine. Weak communication channels in the brain are one consequence of this neuronal deterioration [3]. PD can be defined by stiffness, shakes, difficulty of movement, and issues with body motor function. The symptoms are caused by the slow deterioration of dopaminergic cells in the brain's substantia nigra region. [4]. Even though this disease's source and cure are unknown at this time [5], the availability of numerous medication therapies offers a significant reduction in symptoms, particularly in the early stages, improving patients' life quality and lowering the expected price of the pathology. However, it is commonly accepted that PD results from an overlap of external and biological variables. Research has indicated that more than 89% of people with PD have trouble speaking [6].

Clinical symptoms are usually used to identify Parkinson's disease, yet it has no known cure. Family members, caregivers are all impacted by PD in addition to the individual who has been diagnosed. Each person experiences PD differently, as do its symptoms and rate of progression. To understand it better the underlying causes of the illness, develop more effective treatments, and finally discover a solution, research is still being conducted. In order to improve patient outcomes and start interventions on time, early and correct the identification of PD is crucial. However, because of the delicate nature of the initial symptoms and their potential overlap with other neurological disorders, recognize PD, especially in its initial phases, remains difficult.

- Objective of this study: Analyze the specifics and differences that set PD patients separate from healthy people using speech analysis.
- Role of Machine Learning and AI: For PD tracking and early detection, AI and ML are essential. It is able to diagnose and track diseases by evaluating sensor data, imaging, and clinical records.
- Key Focus Area: To identify the most crucial feature to detect Parkinson. Analyze the effects of medical data which is not balanced.
- Filling the Diagnosis Gap: Utilizing speech attribute as diagnostic tool and aims to increase accuracy through machine learning. The ML will make it easier to detect Parkinson early.
- Research Aim: Deciding which ML model is most effective in early Parkinson's disease detection.

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II. LITERATURE REVIEW

Parkinson's disease is a disorder that affects the nervous system. The use of deep learning and machine learning models for initial identification has been extensively studied. Numerous researches have shown that multiple ML methods are effective in increasing accuracy of diagnoses and making screening procedures possible.

A study by Govindu and Palwe (2023) [7] focused on telemedicine applications in early detection of Parkinson's disease using machine learning. Using the MDVP voice dataset, they assessed Support Vector Machine, Random Forest, K-Nearest Neighbors, and Logistic Regression. Their findings showed the possibility of ML models in separate PD detection, with Random Forest achieving the greatest accuracy of 91.83%.

Gulati et al. (2024) [8] investigated machine learning models for Parkinson's detection using speech features. Random Forest and SVM classifiers were compared in their analysis; SVM achieved 85.71% accuracy, while RF achieved 91.83%. The value of feature selection and model optimization in PD diagnosis is proven by their findings.

Gupta et al. (2023) [9] studied deep learning models for voice-based PD detection. They showed that XGBoost had the highest accuracy of 92.30% using Artificial Neural Networks, XGBoost, and Support Vector Machines. The study highlighted how deep learning models can be useful to distinguish among individuals with Parkinson's disease and a healthy person.

Hussain et al. (2022) evaluated [10] various machine learning techniques for voice-based early PD detection. After applying KNN, SVM, and Logistic Regression, they found that SVM performed better than the other models and achieved 95% accuracy rate. Non-invasive voice biomarkers have been useful in the initial identification of Parkinson's disease.

Rathee et al (2023) [11] carried out an empirical assessment of ML models for speech analysis based PD detection. They applied SVM, Random Forest, Decision Trees, and a new classifier dubbed HyperTab. According to their findings, they achieved the accuracy of 72.29% HyperTab and Random Forest 72.60% which performed the best. The research demonstrated how well speech characteristics and feature engineering work in ML driven PD categorization.

The research demonstrates the efficacy of deep learning and machine learning methods in identifying Parkinson's disease through speech-based analysis. To enhance model performance, accuracy heavily depends on feature selection, dataset balancing, and hyperparameter tuning. Through the analysis of several machine learning classifiers on PD speech datasets, highlighting data imbalance concerns. By improving feature selection procedures, this study builds on previous findings to increase the accuracy of diagnosis.

III. METHODOLOGY

In this study, we use the UCI dataset on speech modulations in Parkinson's patients. Information on the shimmer, jitter, and MDVP of vowel phonations is included in the dataset. Preprocessing, analysis, and visualization of the data are done to ensure that the attributes are well understood. Several models such as RF, LR, NB, DT, KNN, AdaBoost, CatBoost, SVM, Gradient Boosting were used and to get more accurate result hyperparameter tuning was used and gained 95.79% accuracy with CatBoost. Based on frequency fluctuations, models are trained to categorize audio data into two groups: PD and healthy. To train the models we used 80% of the data and to test the models we used 20% of the data, and their accuracy, precision, F1 score, and recall are assessed. To assess how well the model applies to new data, its performance measures are examined. The model's advantages and disadvantages are revealed by this examination. Comparing several machine learning algorithms or iterations of the same method is made possible by model testing. The classifiers were used before and after applying feature selection and data balancing to compare the result.

Fig.2. is the system architecture of Parkinson's Disease prediction using the voice-based biomarkers. It starts with data preprocessing including null values and outliers. Then it split into training and testing in 80-20 ratio. The data is balanced to make sure equal representation. Important features are selected and different ML algorithms are applied to train and test the model. The best model is chosen by comparing their performance and can predict Parkinson's disease or healthy.

A. Dataset Description

There are 195 data points with 10 attributes each in the proposed work. The data set contains a number of biomedical voice measurements obtained from 31 people, in which 23 were with Parkinson disease. We split the dataset in 80-20 ratio. 80% train set, 20% test set. Each of the 195 data points that made up the initial data set for our study reflected patient information pertaining to PD. The UCI Machine Learning repository provided the dataset. [12].

B. Data Preprocess

No values were missing. Due to an imbalance in the target variable (status) suggesting Parkinson's disease, we used SMOTE [13, 14] on the training dataset to balance it. By doing this, the dataset was guaranteed to be complete and prepared for preprocessing.

C. Feature Selection

Using RFE, we enhanced the model's predictive performance and reduced dimensionality [15, 16]. Let S be a subset of the available features and $J(S)$ the model score. Ideally, we would like to find

$$S_{best} = \arg\max_{S} J(S).$$

Since testing every possible combination of features is not feasible, RFE uses a simple backward process. It iteratively trains a model, ranks features by importance, removes the least important one and refits the model until a predefined number of features remains, retaining only the most informative vocal attributes. Top 10 feature were selected and they are as follows ['MDVP:Fo(Hz)', 'MDVP:APQ', 'MDVP:RAP', 'MDVP:Fhi(Hz)', 'Jitter:DDP', 'Shimmer:APQ5', 'NHR', 'spread1', 'D2', 'PPE']. We chose 10 features to balance performance and simplicity, this reduces original feature space while still keeping the main PD related vocal biomarkers and it helps limit overfitting on our small dataset. In addition to using RFE for selecting 10 features, we evaluated feature importance as shown in **Fig.1**. Features such as MDVP:Fo(Hz), Jitter:DDP, Shimmer:APQ5 and NHR are known to reflect vocal irregularities symptoms like pitch instability, amplitude perturbation. These are clinically

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observed symptoms aligning our model with established speech pathology. This supports effectiveness of RFE and enhances the clinical relevance of our finding. **Table I** gives a brief domain specific explanation of each feature, strengthens the clinical relevance and underscore their role in detecting affected from healthy people.

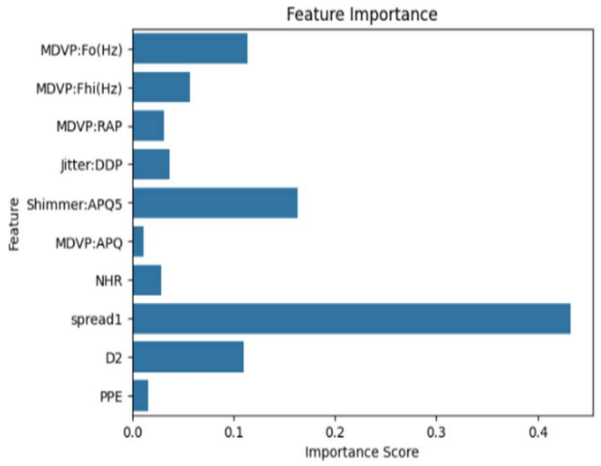


Fig. 1. Feature importance of top 10 selected features using RFE.

TABLE I. DOMAIN SPECIFIC EXPLANATION OF EACH FEATURE

Features	*Description*	*Relevance to PD*
MDVP:Fo(Hz)	Represents average vocal fundamental frequency.	Develop monotonic speech because their voice muscles are affected.
MDVP:Fhi(Hz)	Maximum fundamental frequency.	Reduce vocal modulation capacity.
MDVP:RAP	Relative amplitude perturbation	Cause impaired motor control of vocal muscles.
Jitter:DDP	Analyze pitch variations of three continuous voice cycles.	Measures pitch instability over multiple speech cycles.
Shimmer:APQ5	Amplitude perturbation quotient on 5 point.	Measures voice instability.
MDVP:APQ	Quantifies amplitude perturbation.	Unstable voice volume.
NHR	Noise to harmonic ratio.	Indicates breathy or hoarse voice.
spread1	Nonlinear dynamic measure of signal.	Captures chaotic vocal fold vibrations.
D2	Correlation dimension.	Measures complexity of vocal fold vibration pattern.
PPE	Pitch period entropy.	Quantifies pitch variation patterns.

D. Training and Testing

Machine learning algorithms like CatBoost, DT, AdaBoost, NB, KNN, LR, RF, SVM, and Gradient Boosting use training data sets to teach the model to do a range of tasks. We split the dataset in an 80-20 ratio. For training we used 80% data and for testing 20%. Detailed features from the training set are extracted in order to train the model. This is why these structures are included in the prototype. The training set is then appropriately titled with the model to benefit from the components. Using hyperparameter tuning helps optimize ML model, which improved accuracy. For hyperparameter tuning, grid search with five-fold cross-validation is applied.

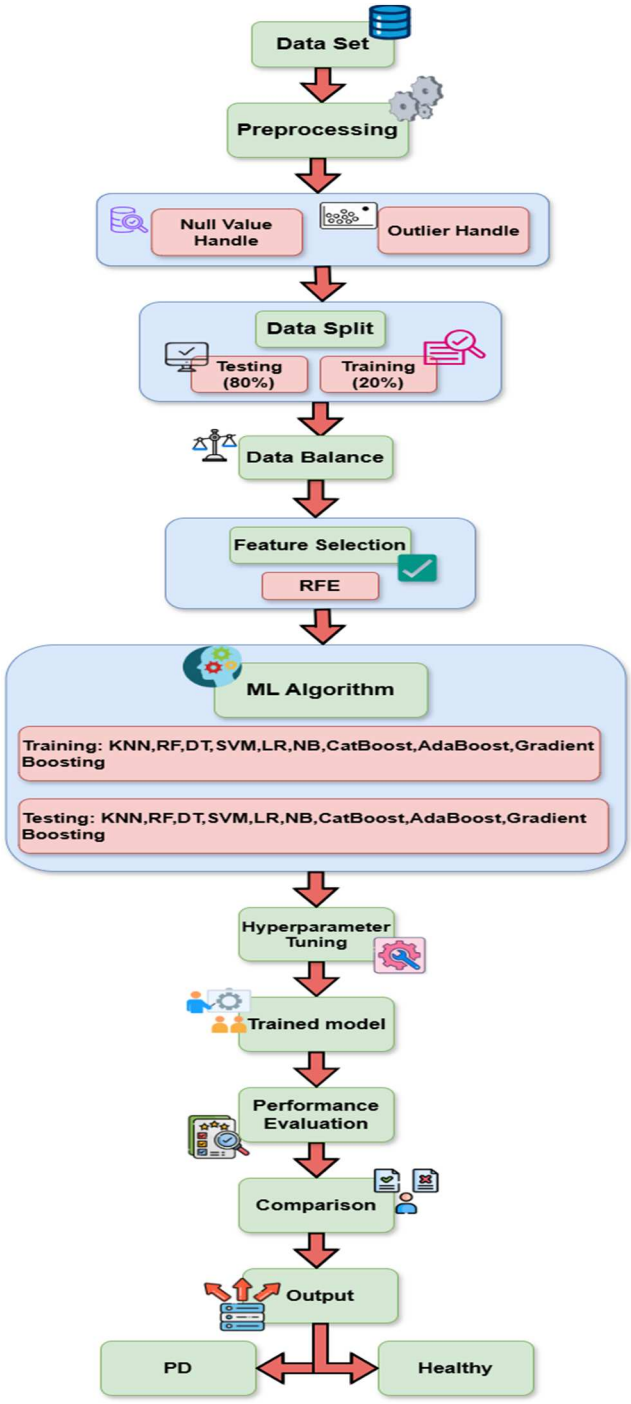


Fig. 2. System Architecture of Parkinson's Disease Prediction.

IV. RESULT AND DISCUSSION

AdaBoost, CatBoost, RF, SVM, DT, NB, KNN, LR, and Gradient Boosting are a few of the well known machine learning classifiers that were used to train the dataset. To determine each model's advantages and disadvantages, its precision, accuracy, recall, F1-score, and support were assessed. The model showed the accuracy of the classifiers before and after feature selection and data balancing to get a clear understanding. Through these steps the accuracy was improved. Our feature selection method RFE, helped simpler models, was neutral for tree ensembles, and initially degraded performance for distance and kernel based methods. To optimize model performance and enhance accuracy, we applied hyperparameter tuning using grid search which

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identified CatBoost as the best performing classifier. It achieved 95.79% accuracy, strong ROC performance and fast inference by capturing non-linear interactions among voice features, so we present it as a practical best choice.

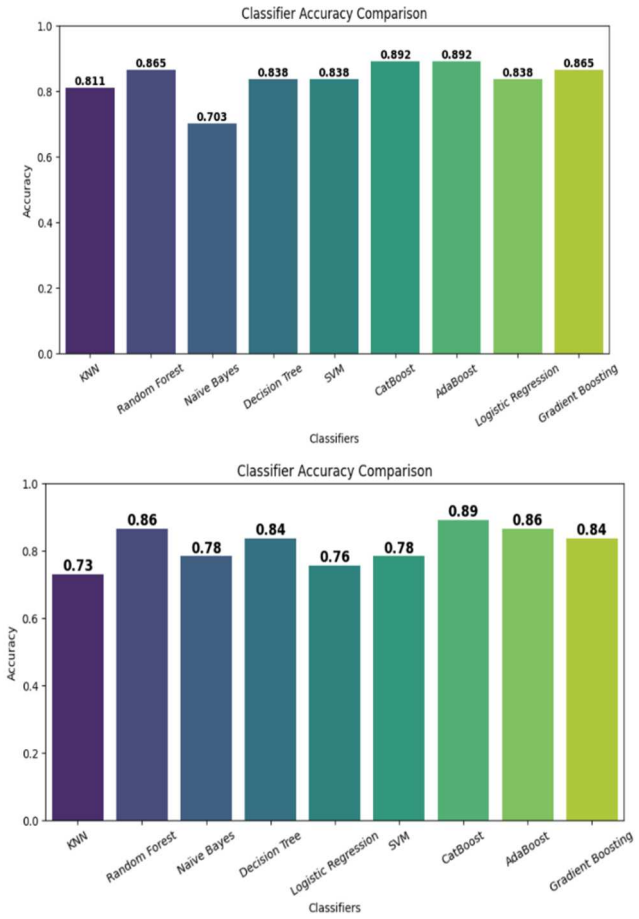


Fig. 3. Comparison between accuracy before and after feature selection.

Fig.3. is representing pre and post results after using feature selection of different classifiers. The highest accuracy was given by CatBoost and AdaBoost which was 89% before feature selection. However, we applied feature selection to reduce overfitting which displays slightly lower accuracy range but it improves efficiency and achieved accuracy of 89% by CatBoost. Feature selection may have reduced the overall accuracy but it helped simplifying the model by focusing on the most important features with more practical and robust approach for real world application.

TABLE II. PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS WITHOUT FEATURE SELECTION FOR PARKINSON'S DISEASE PREDICTION

Index	*Model*	*Accuracy (%)*	*Precision (%)*	*Recall (%)*	*F1-Score (%)*
1	KNN	81.08	83.81	81.08	81.9
2	Random Forest	86.49	86.09	86.49	86.21
3	Naïve Bayes	70.27	82.5	70.27	72.38
4	Decision Tree	83.78	83.78	83.78	83.78
5	SVM	83.78	82.96	83.78	83.05
6	CatBoost	89.19	89.19	89.19	89.19
7	AdaBoost	89.19	89.19	89.19	89.19
8	Logistic Regression	83.78	83.78	83.78	83.78
9	Gradient Boosting	86.49	87.1	86.49	86.72

TABLE III. PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS WITH FEATURE SELECTION FOR PARKINSON'S DISEASE PREDICTION

Index	*Model*	*Accuracy (%)*	*Precision (%)*	*Recall (%)*	*F1-Score (%)*
1	KNN	72.97	80.15	72.97	74.73
2	Random Forest	86.49	87.1	86.49	86.72
3	Naïve Bayes	78.38	85.21	78.38	79.79
4	Decision Tree	83.78	85.33	83.78	84.29
5	SVM	75.66	78.77	75.66	76.75
6	CatBoost	78.38	76.96	78.38	77.4
7	AdaBoost	89.19	89.19	89.19	89.19
8	Logistic Regression	86.49	91.31	86.49	87.29
9	Gradient Boosting	83.78	85.33	83.78	84.29

A through comparison of model performance without and with feature selection are shown in **Table II** and **Table III**. The table is based on the model's accuracy, precision, recall and F1-score. Accuracy ranging from 72.27% (Naïve Bayes) to 89.19% (CatBoost and AdaBoost) was achieved without selecting feature. Accuracy dropped slightly for some models after using feature selection ranging from 72.97% (KNN) to 89.19% (AdaBoost). Naïve Bayes improves from 70.27% to 78.38, suggesting feature selection helped this model. AdaBoost remains the same at 89.19%. Feature selection has it's mixed impact for the classifiers.

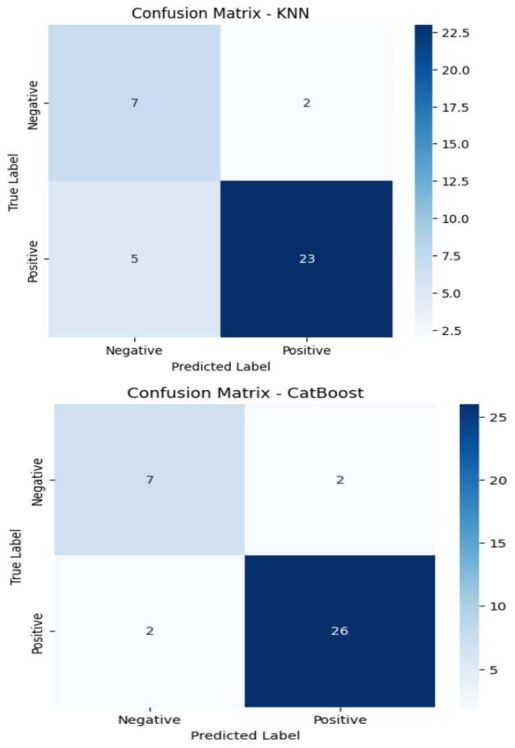


Fig. 4. Confusion matrix before feature selection.

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In **Fig. 4, 5** and **6** the confusion matrix represents before and after feature selection and after hyperparameter tuning. To determine the accuracy and correctness of the models, the most user friendly matrix is a confusion matrix. It explains how well categorization models work when the truth values are known beforehand. In **Fig.7** shows ROC curves of different tuned ML classifiers.

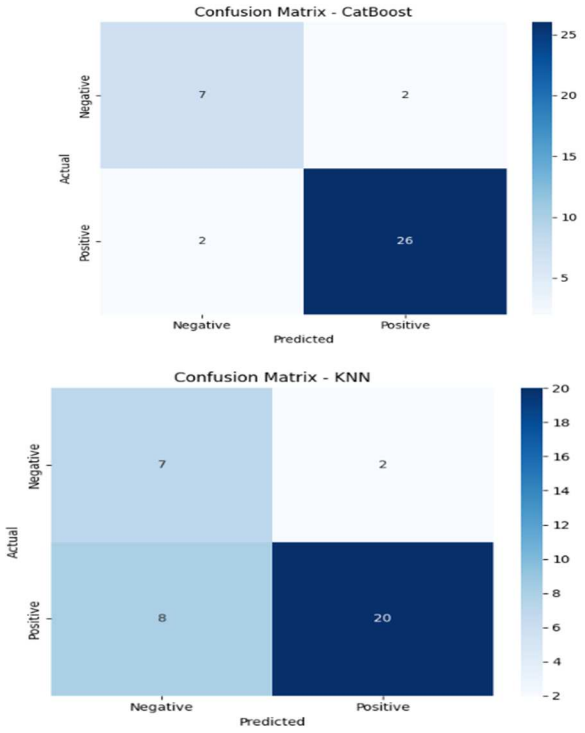


Fig. 5. Confusion matrix after feature selection.

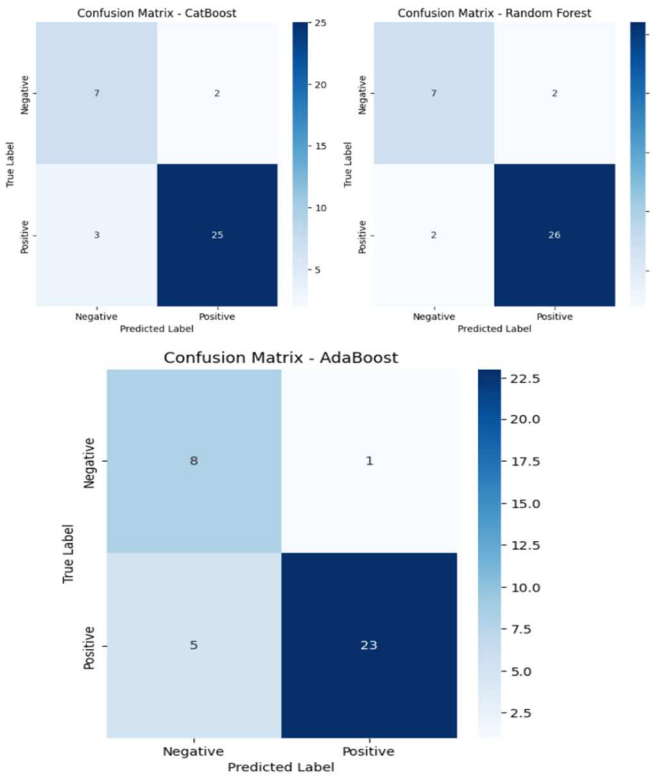


Fig. 6. Confusion matrix after hyperparameter tuning.

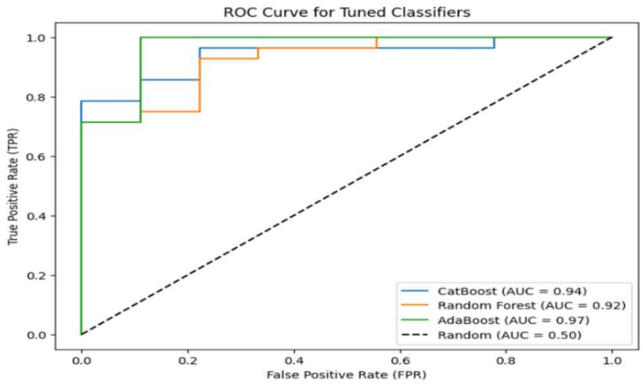


Fig. 7. ROC curves of different ML classifiers.

A. *Computational Efficiency*

In addition to evaluate accuracy and other classification, we compared training time and inference time of all classifiers. Our findings show that CatBoost achieved highest accuracy but required longest training time (8.04s) but it has the fastest inference speed (0.0011s) make it a strong candidate for real time prediction. **Table IV** gives brief comparison of training time and inference speed of each classifier. These findings represent that while Catboost is computationally rigorous during training but its efficient and accurate in prediction.

TABLE IV. COMPUTATIONAL COMPARISON OF EACH CLASSIFIER

Classifier	*Training Time (s)*	*Inference Speed (s)*
KNN	0.008859	0.019736
Random Forest	0.296779	0.008045
Naïve Bayes	0.001914	0.001224
Decision Tree	0.003038	0.001163
Logistic Regression	0.017049	0.002514
SVM	0.024709	0.002490
CatBoost	8.038995	0.001134
AdaBoost	0.198077	0.020050
Gradient Boosting	0.548219	0.001775

B. *Statistical Backing*

The performance differences between AdaBoost, Random Forest, and CatBoost are shown with confidence intervals in **Table V**. Although Random Forest exhibits somewhat better mean differences over CatBoost and AdaBoost, all comparisons provide confidence intervals crossing zero, suggesting no statistically meaningful superiority.

TABLE V. CONFIDIENCE INTERVAL FOR PERFORMANCE DIFFERENCE

Comparison	***Mean Difference***	***95% CI Lower***	***95% CI Upper***	***Interpretation***
CatBoost vs Random Forest	-0.027	-0.122	0.068	Random Forest slightly better, but CI includes 0
CatBoost vs AdaBoost	0.027	-0.040	0.094	CatBoost slightly better, but CI includes 0
Random Forest vs AdaBoost	0.054	-0.023	0.131	Random Forest better, but CI includes 0

Based on the provided data set, the different machine learning methods for Parkinson's disease identification demonstrate that CatBoost has attained its F1-Score, accuracy, precision, and recall. **Table VI** gives a brief performance comparison of existing study and proposed model.

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TABLE VI. COMPARISON OF MACHINE LEARNING MODELS FOR PARKINSON'S DISEASE PREDICTION ACROSS EXISTING STUDIES

No.	*Author*	*Method*	*Accuracy*
1.	Govindu et al. [7]	Random Forest	91.83%
2.	Gupta et al. [9]	XGBoost	92.30%
3.	Hussain et al. [10]	Support Vector Machine	95%
4.	Gulati et al. [8]	Random Forest	91.83%
5.	Rathee et al. [11]	Random Forest	72.60%
6.	Proposed Model	Catboost	95.79%

Our proposed model is trained and tested with vocal biomarkers of PD patients. The model focused on optimized feature selection which achieved an accuracy of 95.79% using CatBoost, which outperforms other models and existing studies. Recursive Feature Elimination select the top 10 features, reducing dimensionality and focuses on most relevant attributes. Data was also balanced in this study to ensuring equal representation of PD and healthy cases. Hyperparameter tuning with grid search and 5-fold cross validation also used to optimize model's performance and improve the efficiency and accuracy of detecting Parkinson disease.

V. CONTRIBUTION AND LIMITATION

A. Contributions

Although previous research has used machine learning model on the same dataset, our study makes several contributions. The study demonstrated the efficiency in predicting Parkinson's disease using vocal biomarkers. We applied nine machine learning classifiers and also included less studied methods like CatBoost and Gradient Boosting. We combined hyperparameter tuning to improve performance, SMOTE for data balance and RFE for effective feature selection. It is a systematic method that was often absent in earlier studies. We exceed previous studies in the literature by applying CatBoost with an accuracy of 95.79%. Additionally, we present feature selection pre and post results which provide a view how dimensionality reduction affects a classifier's performance. Also included computational efficiency, domain specific explanation of each feature for model's clinical relevance.

B. Limitation

This study uses a small dataset (195 samples) sourced from a single repository. While dataset provides voice-based biomarkers of PD, it might not accurately reflect differences seen in real world clinical contexts. Factors as recording conditions, device type etc. difference can impact on model's generalizability. In future work we want to evaluate the model with larger and multisource dataset to ensure robustness and improve clinical effectiveness.

VI. CONCLUSION

This study showed how well ML models use speech biomarkers to predict PD. The study tested various ML algorithms on the dataset. This research highlighted the significance of feature selection and data balancing to enhance performance. When Parkinson's illness is classified using vowel phonation data, the accuracy is 95.79% using CatBoost. Because each attribute in the dataset is given equal weight, these model produces optimal results. The findings suggest that CatBoost combined with feature selection and hyperparameter tuning is a robust approach for early PD. For future work, we will Cross-validate the model on larger, multi-source datasets to improve robustness.

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